Introduction: Description – Dataset

In the context of this exercise, you will be analyzing a recurring survey for a subset of outlets and products for a given time period. The purpose of the survey is to measure different execution metrics in the outlet, for example, if a cooler is present, if there are promotional materials, and if specific products are available to be purchased by consumers at the outlet. The later will be the focus of this exercise. The survey is taken at different time intervals for each outlet, based on when the outlet is visited, and contains information if a given product was available for purchase or not on a particular date. Your task is to analyze this dataset and provide insights and predictions regarding the product availability for the target products.

You are provided with a dataset (data.csv) that contains the following columns:

Column	Description
outlet	The identifier for the outlet where the survey was conducted
date	The date of the survey
product	The identifier of the product
available	Binary indicator whether the product was available (1) or not available (0) on the given date and outlet
outlet_type	The type/channel of the outlet
product_type	The type of the product
sales_performance_indicator	Indicator ranging from 1 to 5, 1 being the highest in terms of sales performance and 5 being the lowest
outlet_volume_potentiality	Potentiality of outlet in terms of volume, compared to other similar outlets. "High" means that there are other similar outlets that have achieved much higher volume, while "Zero" means that the performance is at its maximum. This relates to sales across all product categories
outlet_category_volume_potentiality	Similar to above, but for the specific product category
cooler_presence	Binary indicator whether a cooler is present in the outlet (1), or not (0)
outlet_satisfaction	Level of satisfaction of outlet owner with the services (10=Most satisfied, 1=Least satisfied)
price_range_indicator	Indicator regarding the prices of the outlet, as seen through various websites e.g. google, tripadvisor. (4) means very expensive/premium, while (1) means very affordable
consumer_sentiment	Score from 0 to 100 indicated how the consumers feel about the outlet, based on social media reviews (100=Extremely satisfied, 0=Least satisfied)

Column	Description
popularity_with_consumers	Similar to above, taking into account also the number of reviews, in order to more closely approximate the popularity of the outlet
consumer_disposable_income_index	Disposable income index regarding people who live around the outlet. 100 is the mean of the country population, and values above or below indicate more expensive or affordable areas
consumer_spending_per_capita_on_food_beverages	Yearly spending in euros on food and beverages per capita for people living around the outlet
surrounding_traffic	Average monthly foot traffic (people passing) around the proximity of the outlet
touristic_point_of_interest	Number of touristic point of interest (museums, monuments etc.) around the proximity of the outlet
number_of_competing_outlets	Number of competing outlets (same outlet_type) around the given outlet

Part 1: Exploration & Insights

Explore the data and provide insights into any patterns or trends in product availability. Are there particular outlets or dates where product availability for target products is consistently high or low? Are there any interesting elements among the outlets attributes that were provided? Suggest potential factors or reasons for low availability and propose strategies for improvement (could be outside the ones given in the dataset).

Part 2: Predictive Modeling

Using an appropriate machine learning algorithm, develop an approach that will predict what is the probability that an outlet with given characteristics is going to have the target product available. There is no constraint regarding the modeling approach you choose to follow.

Discuss the strengths and weaknesses of the modeling approach you chose, the model performance and suggest possible improvements.

Part 3: Targeting

[No coding is required for this part]

The dataset that was provided contains a small subset of the outlets and products. In reality there are hundreds of products and hundreds of thousands of outlets. Today, based on simple grouping criteria (example all cafes), a given product is set as target for the outlet. For example, product_1 is a target for cafes and column "available" indicates if the target is met or not.

Imagine you are tasked with automating this targeting and develop an approach that will automatically identify which outlets should be matched with which products as targets. How would you approach it as a problem, what are the key questions you would ask from business stakeholders, what are the business and technical aspects that should be considered, in order to develop this as an end-to-end solution?

Conclusion: Deliverables

You are free to work with your preferred environment, using python/pyspark. As part of the deliverable, a presentation (.pptx) in English is expected showcasing your findings to business stakeholders (that may include non-technically savvy people). The document should be accompanied by a zip file containing the source code (in .py, .ipynb) appropriately documented and annotated.